

Exercises from Ch. 9: Algebraic Spaces, Lang

Matthew Gergley

Last Updated: August 21, 2026

Disclaimer: The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

Exercises

Spec of a ring

Exercise 0.1. (Exercise 14)

Let A be a commutative ring. Define $\text{spec}(A)$ to be **connected** if $\text{spec}(A)$ is not the union of two disjoint non-empty closed sets (or equivalently, $\text{spec}(A)$ is not the union of two disjoint, non-empty open sets).

- a) Suppose that there are idempotents e_1, e_2 in A (that is $e_1^2 = e_1$ and $e_2^2 = e_2$), $\neq 0, 1$, such that $e_1 e_2 = 0$ and $e_1 + e_2 = 1$. Show that $\text{spec}(A)$ is not connected.
- b) Conversely, if $\text{spec}(A)$ is not connected, show that there exist idempotents as in part (a).

In either case, the existence of the idempotents is equivalent with the fact that the ring A is a product of two non-zero rings, $A = A_1 \times A_2$.

Proof. a) Assume that $0, 1 \neq e_1, e_2 \in A$ are idempotents such that $e_1 e_2 = 0$ and $e_1 + e_2 = 1$. Let $\mathfrak{p} \in \text{spec}(A)$. Since $0 \in \mathfrak{p}$, $e_1 e_2 \in \mathfrak{p}$ which means that $e_1 \in \mathfrak{p}$ or $e_2 \in \mathfrak{p}$. Furthermore, since $1 \notin \mathfrak{p}$ (otherwise $\mathfrak{p} = A$) we cannot have both e_1 and e_2 in $\mathfrak{p}$ by the relation $e_1 + e_2 = 1$. Thus, consider the subsets of $\text{spec}(A)$ defined as

$$\mathcal{L}((e_1)) = \{\mathfrak{p} \in \text{spec}(A) : (e_1) \subseteq \mathfrak{p}\} \quad \text{and} \quad \mathcal{L}((e_2)) = \{\mathfrak{p} \in \text{spec}(A) : (e_2) \subseteq \mathfrak{p}\}.$$

Note that in particular, every prime ideal in $\mathcal{L}((e_1))$ contains e_1 and every prime ideal in $\mathcal{L}((e_2))$ contains e_2 . Hence, $\mathcal{L}((e_1)), \mathcal{L}((e_2)) \subseteq \text{spec}(A)$ are closed since $(e_1), (e_2) \subseteq A$ are ideals. From this and the relation $e_1 + e_2 = 1$,

$$\mathcal{L}((e_1)) \cap \mathcal{L}((e_2)) = \emptyset.$$

Now, if $\mathcal{L}((e_1)) = \emptyset$, then $e_1 \notin \mathfrak{p}$ for all $\mathfrak{p} \in \text{spec}(A)$. Hence, $e_1 \in A$ is a unit, so

$$e_1^2 = e_1 \Rightarrow e_1^{-1} e_1^2 = 1 \Rightarrow e_1 = 1$$

which contradicts the assumption that $e_1 \neq 1$. Thus, $\mathcal{L}((e_1)) \neq \emptyset$. By a similar argument, we have $\mathcal{L}((e_2)) \neq \emptyset$ as well.

Let $\mathfrak{p} \in \text{spec}(A)$ be arbitrary. If $e_1 \in \mathfrak{p}$, then $\mathfrak{p} \in \mathcal{L}((e_1))$ and likewise if $e_2 \in \mathfrak{p}$, then $\mathfrak{p} \in \mathcal{L}((e_2))$. Hence, every point in $\text{spec}(A)$ belongs to either $\mathcal{L}((e_1))$ or $\mathcal{L}((e_2))$. Thus,

$$\mathcal{L}((e_1)) \cup \mathcal{L}((e_2)) = \text{spec}(A).$$

Therefore, $\text{spec}(A)$ is not connected.

b) Assume that $\text{spec}(A)$ is not connected and write

$$\text{spec}(A) = X \cup Y$$

where X, Y are closed subsets such that $X, Y \neq \emptyset$ and $X \cap Y = \emptyset$. Thus, there exists ideals $I, J \subseteq A$ such that $I \subseteq \mathfrak{p}$ and $J \subseteq \mathfrak{q}$ for all $\mathfrak{p} \in X$ and $\mathfrak{q} \in Y$, respectively. Hence,

$$X = \mathcal{L}(I) = \{\mathfrak{p} \in \text{spec}(A) : I \subseteq \mathfrak{p}\} \quad \text{and} \quad Y = \mathcal{L}(J) = \{\mathfrak{p} \in \text{spec}(A) : J \subseteq \mathfrak{p}\}.$$

Thus, by Proposition 5.1,

$$X \cap Y = \emptyset \Rightarrow \mathcal{L}(I) \cap \mathcal{L}(J) = \mathcal{L}(I + J) = \emptyset$$

and

$$X \cup Y = \text{spec}(A) \Rightarrow \mathcal{L}(I) \cup \mathcal{L}(J) = \mathcal{L}(IJ) = \text{spec}(A).$$

Then $I + J \not\subseteq \mathfrak{p}$ and $IJ \subseteq \mathfrak{p}$ for all $\mathfrak{p} \in \text{spec}(A)$. This means that for an arbitrary $x + y \in I + J$, we have $x + y \notin \mathfrak{p}$ for all $\mathfrak{p} \in \text{spec}(A)$ and thus $x + y$ is unit. In particular, $1 \in I + J$, so $I + J = A$. The relation $IJ \subseteq \mathfrak{p}$ for every point of $\text{spec}(A)$ implies that $xy \in \mathfrak{p}$ for every $\mathfrak{p} \in \text{spec}(A)$. Hence,

$$xy \in \bigcap_{\mathfrak{p} \in \text{spec}(A)} \mathfrak{p} = \sqrt{(0)} = \mathcal{N}_A = \{a \in A : a^n = 0 \text{ for some } n \in \mathbb{Z}_{\geq 1}\}.$$

Thus, there exists an integer $n \geq 1$ such that $(xy)^n = 0$.

Choose $a \in I$ and $b \in J$ such that $a + b = 1$ and let $n \geq 1$ satisfy $(ab)^n = 0$. Since $b = 1 - a$,

$$-ab = -a(1 - a) = a^2 - a \in IJ \quad (\text{neg. sign makes computation "cleaner"})$$

Hence, $(a^2 - a)^n = 0$.

Consider the polynomial

$$f(X) = \sum_{k=n}^{2n-1} \binom{2n-1}{k} X^k (1-X)^{2n-1-k} \in A[X]$$

and define $e = f(a) = \sum_{k=n}^{2n-1} \binom{2n-1}{k} a^k (1-a)^{2n-1-k}$. By the binomial theorem, we may write

$$1 = (X + (1-X))^{2n-1} = \sum_{k=0}^{2n-1} \binom{2n-1}{k} X^k (1-X)^{2n-1-k}$$

and thus

$$1 - f(X) = \sum_{k=0}^{n-1} \binom{2n-1}{k} X^k (1-X)^{2n-1-k}.$$

Since $(1-X)^n$ divides every term of the sum on the right-hand side,

$$f(X) \equiv 1 \pmod{(1-X)^n}$$

and $X^n \mid f(X)$, i.e.,

$$f(X) \equiv 0 \pmod{X^n}.$$

Consequently,

$$f(X)^2 - f(X) = f(X)(f(X) - 1)$$

is divisible by both X^n and $(1 - X)^n$ with X^n and $(1 - X)^n$ relatively prime, so in particular

$$X^n(1 - X)^n \mid f(X)^2 - f(X) \Rightarrow f(X)^2 - f(X) = X^n(1 - X)^n h(X), \quad h(X) \in A[X].$$

Evaluating at $X = a$,

$$\begin{aligned} f(a)^2 - f(a) &= a^n(1 - a)^n h(a) \\ &= a^n b^n h(a) = (ab)^n h(a) = 0. \end{aligned}$$

Therefore, $f(a)^2 = f(a)$, that is, $e^2 = e$. We show that $e \neq 0, 1$. Every term in $e = f(a)$ is divisible by a^n ($a \in I$), so $e \in I$. Since $I \subseteq A$ is proper, it follows that $e \neq 1$. Similarly, every term of $1 - e = 1 - f(a)$ is divisible by $(1 - a)^n = b^n$ ($b \in J$), so $1 - e \in J$. Again, since $J \subseteq A$ is proper, we have $1 - e \neq 1$ which implies that $e \neq 0$. Hence,

$$e^2 = e \quad \text{and} \quad e \neq 0, 1,$$

so e is an idempotent. Now we construct e_1 and e_2 . Define

$$e_1 = e \quad \text{and} \quad e_2 = 1 - e.$$

We show that these are idempotents and satisfy the relations stated in (a). Clearly, $e_1, e_2 \neq 0, 1$ since $e \neq 0, 1$. Furthermore,

$$e_1^2 = e^2 = e = e_1 \quad \text{and} \quad e_2^2 = (1 - e)^2 = 1 - 2e + e^2 = 1 - e = e_2.$$

Finally,

$$e_1 + e_2 = e + (1 - e) = 1 \quad \text{and} \quad e_1 e_2 = e(1 - e) = -(e^2 - e) = 0.$$

Thus, we have the desired idempotents e_1 and e_2 . □

Exercise 0.2. (Exercise 15)

Prove that the Zariski topology is **compact**, in other words: let $\{U_i\}_{i \in I}$ be a family of open sets such that

$$\bigcup_i U_i = \text{spec}(A).$$

Show that there is a finite number of open sets $U_{i_1}, \dots, U_{i_n}$ whose union is $\text{spec}(A)$. [*Hint*: Use closed sets, and use the fact that if a sum of ideals is the unit ideal, then 1 can be written as a finite sum of elements.]

Proof. Let $\{U_i\}_{i \in I}$ be a family of open sets of $\text{spec}(A)$ such that

$$\bigcup_i U_i = \text{spec}(A).$$

Define

$$V_i := \text{spec}(A) \setminus U_i \subseteq \text{spec}(A)$$

which are closed. Hence, there exist ideals $\mathfrak{a}_i \subseteq A$ such that $V_i = \mathcal{L}(\mathfrak{a}_i)$. Taking the complement of $\text{spec}(A)$,

$$\begin{aligned}
\emptyset = \text{spec}(A) \setminus \text{spec}(A) &= \text{spec}(A) \setminus \bigcup_i U_i \\
&= \bigcap_i (\text{spec}(A) \setminus U_i) \quad (\text{By DeMorgan's Law}) \\
&= \bigcap_i V_i \\
&= \bigcap_i \mathcal{L}(\mathfrak{a}_i) = \mathcal{L}\left(\sum_i \mathfrak{a}_i\right) \subseteq \text{spec}(A)
\end{aligned}$$

Note that if $\mathcal{L}(\mathfrak{a}) = \emptyset$ for some ideal $\mathfrak{a} \subseteq A$, then $\mathfrak{a} = (1)$. If not, i.e. $\mathfrak{a} \neq (1)$, then $\mathfrak{a} \subseteq A$ is a proper ideal and thus is contained in a maximal ideal (prime), so $\mathcal{L}(\mathfrak{a}) \neq \emptyset$. Thus, since $\mathcal{L}(\mathfrak{a}) = \emptyset = \mathcal{L}((1))$, we have $\sum_i \mathfrak{a}_i = (1)$. Thus, we may write

$$1 = \sum_{k=1}^n x_{i_k}, \quad x_{i_k} \in \mathfrak{a}_{i_k}$$

for $i_1, \dots, i_n \in I$. We want to show that

$$\bigcup_{k=1}^n U_{i_k} = \text{spec}(A).$$

By way of contradiction, suppose there exists a point $\mathfrak{p} \in \text{spec}(A)$ such that

$$\mathfrak{p} \notin \bigcup_{k=1}^n U_{i_k},$$

i.e. $\mathfrak{p} \notin U_{i_k}$ for all $1 \leq k \leq n$. Hence,

$$\mathfrak{p} \in \text{spec}(A) \setminus \left(\bigcup_{k=1}^n U_{i_k}\right) = \bigcap_{k=1}^n V_{i_k} = \mathcal{L}\left(\sum_{k=1}^n \mathfrak{a}_{i_k}\right).$$

Then $\sum_{k=1}^n \mathfrak{a}_{i_k} \subseteq \mathfrak{p}$. Since $\mathfrak{a}_{i_k} \subseteq \sum_{j=1}^n \mathfrak{a}_{i_j} \subseteq \mathfrak{p}$ and $x_{i_k} \in \mathfrak{a}_{i_k}$, we have $x_{i_k} \in \mathfrak{p}$ for all $1 \leq k \leq n$. Thus,

$$\sum_{k=1}^n \mathfrak{a}_{i_k} = 1 \in \mathfrak{p} \quad (\mathfrak{p} \text{ is closed under finite sums})$$

which is impossible since $\mathfrak{p} \subseteq A$ is prime (proper). Contradiction!

Therefore,

$$\bigcup_{k=1}^n U_{i_k} = \text{spec}(A),$$

and thus the Zariski topology is compact. □